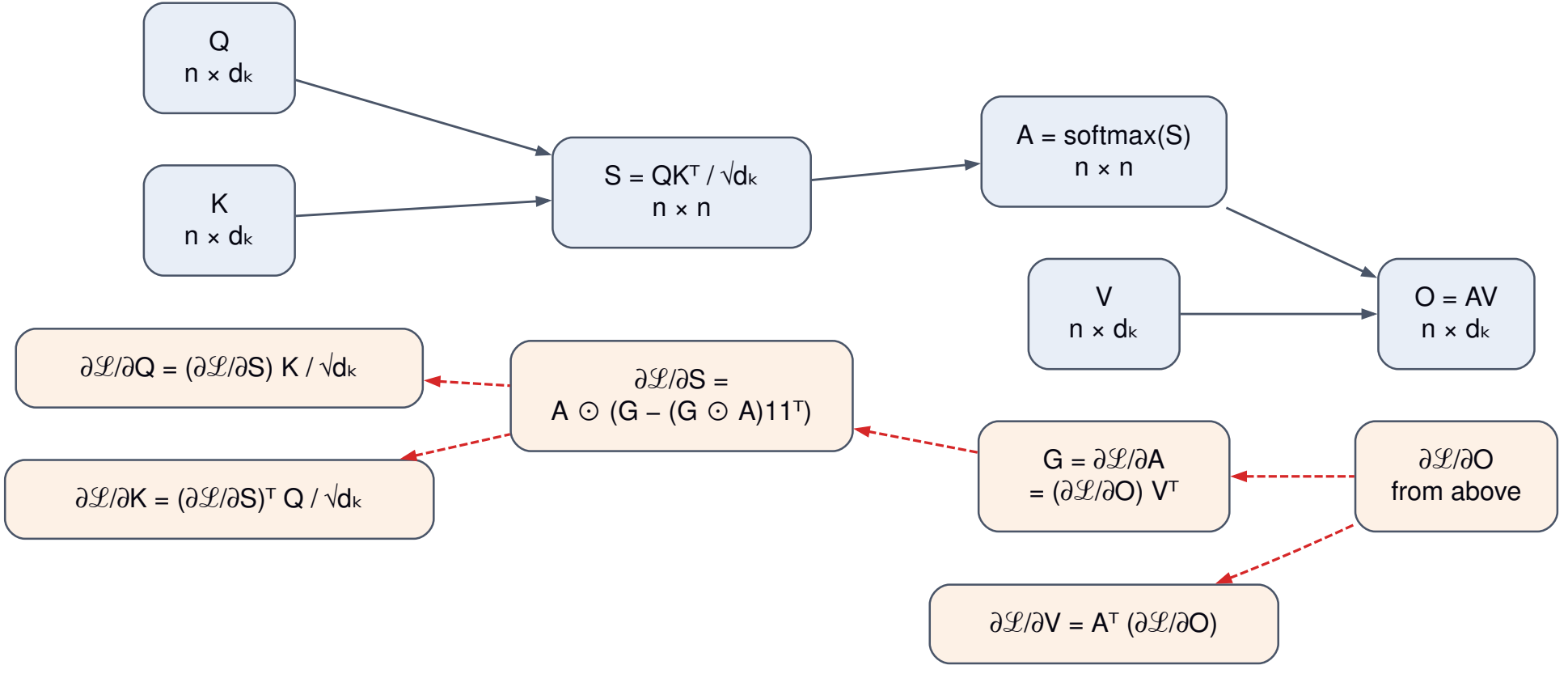
Q
n × dₖ
K
n × dₖ
S = QKᵀ / √dₖ
n × n
A = softmax(S)
n × n
V
n × dₖ
O = AV
n × dₖ
∂ℒ/∂Q = (∂ℒ/∂S) K / √dₖ
∂ℒ/∂K = (∂ℒ/∂S)ᵀ Q / √dₖ
∂ℒ/∂S =
A ⊙ (G – (G ⊙ A)11ᵀ)
G = ∂ℒ/∂A
= (∂ℒ/∂O) Vᵀ
∂ℒ/∂O
from above
∂ℒ/∂V = Aᵀ (∂ℒ/∂O)